Breast Cancer Prediction Using Logistic Regression

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GithubLink: https://github.com/IswaryaLaharikola/breast-cancer-logistic-regression

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1. Introduction

Modern data-driven systems now rely heavily on machine learning, which allows computers to recognize patterns in data and generate accurate predictions without the need for explicit programming (Goodfellow et al., 2016). Assigning input data into predetermined categories is the aim of classification, one of the most popular machine learning tasks (Han et al., 2011). This is especially crucial in fields where making precise decisions is essential, like cybersecurity, healthcare, and finance.

Classification models are frequently employed in the field of medical diagnosis to forecast the presence of a disease based on patient data. Treatment results and survival rates can be greatly enhanced by early identification of diseases like cancer. As a result, trustworthy and comprehensible machine learning models are crucial.

Logistic regression, a basic supervised learning approach for binary classification problems, is the main topic of this lesson (Hosmer et al., 2013). Logistic regression is nonetheless very effective despite its simplicity, particularly when there is a roughly linear connection between the target variable and the input features (James et al., 2021).

This tutorial's goal is to show how Logistic Regression may be used using the Breast Cancer Wisconsin dataset (Wolberg et al., 1995), a real-world dataset. Data preprocessing, model training, assessment, and result interpretation are all covered in detail throughout the lesson. To improve comprehension, visual aids like ROC curves and confusion matrices are also employed

2. Overview of Logistic Regression

For binary classification tasks, where the output variable might have one of two possible values, usually denoted as 0 or 1, logistic regression is a supervised machine learning approach (Hosmer et al., 2013). Logistic regression forecasts probabilities that are linked to class labels, in contrast to linear regression, which forecasts continuous values.

The sigmoid (logistic) function, which converts a linear combination of input data into a probability value between 0 and 1, is the fundamental idea behind logistic regression (James et al., 2021). This guarantees that forecasts stay within a legitimate probability range. To assess the likelihood of belonging to a specific class, the model uses the sigmoid transformation after computing a weighted sum of the input features. The result is then classified using a threshold, typically 0.5.

The interpretability of logistic regression is one of its main benefits. Every attribute has a corresponding coefficient that shows how much of an impact it has on the prediction. A class's likelihood is increased by positive coefficients and decreased by negative ones.

But according to Bishop (2006), logistic regression assumes that the independent factors and the dependent variable's log-odds have a linear connection. In other words, it works best with linearly separable data.

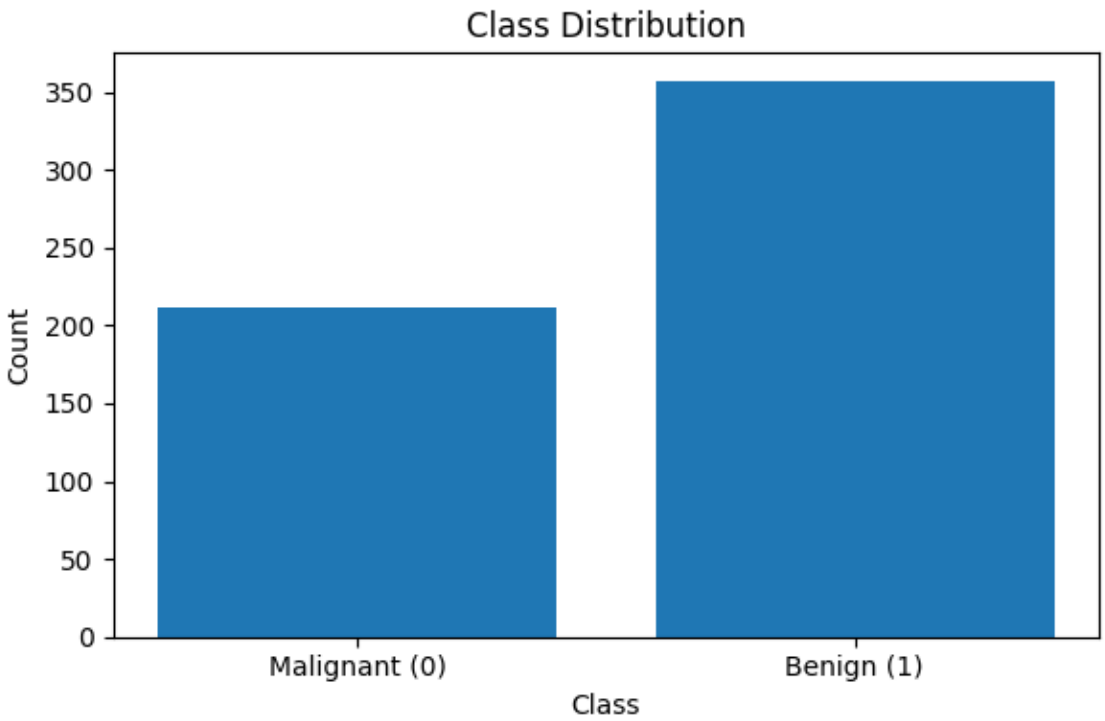
Furthermore, feature scaling is frequently necessary to provide reliable and effective training (Kuhn and Johnson, 2013).

Despite these drawbacks, logistic regression is frequently employed because of its ease of use, effectiveness, and robust performance on a wide range of real-world issues.

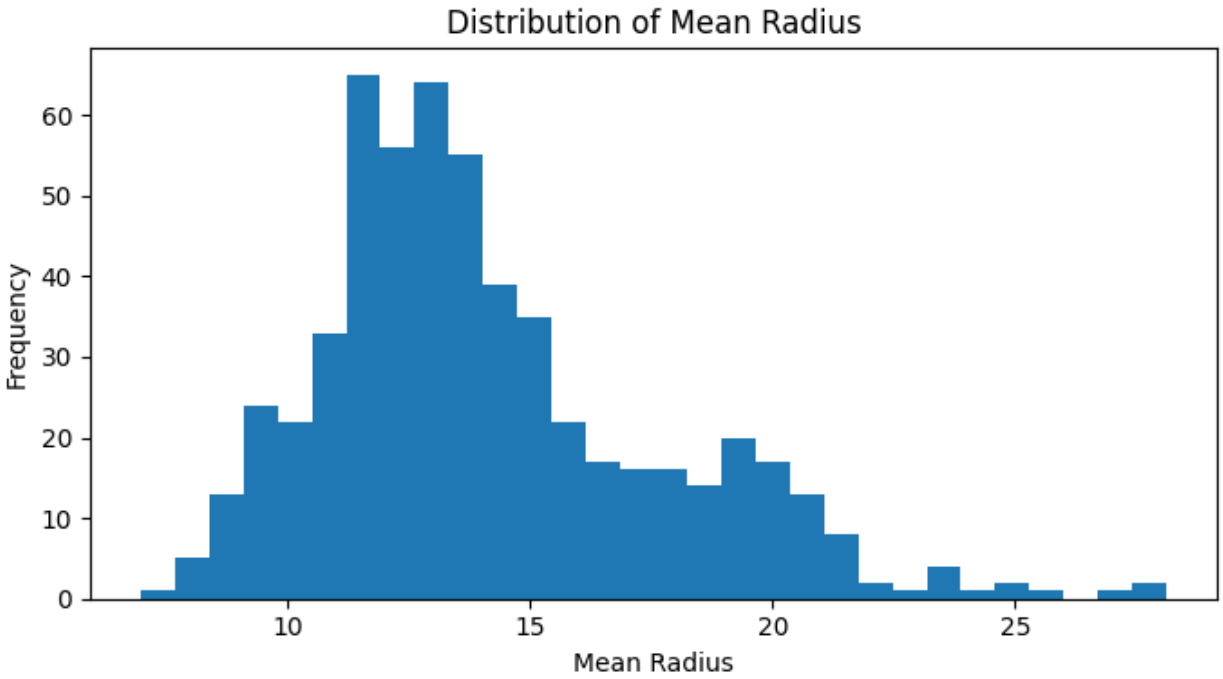
3. Dataset Description

The Breast Cancer Wisconsin dataset, which is frequently utilized for binary classification problems in machine learning, is the dataset utilized in this lesson (Wolberg et al., 1995). It includes measures taken from digital pictures of breast tumor cells.

Thirty numerical features, including radius, texture, perimeter, area, and smoothness, are included in the dataset, which comprises 569 samples. Every sample has a label indicating whether it is benign (1) or malignant (0).

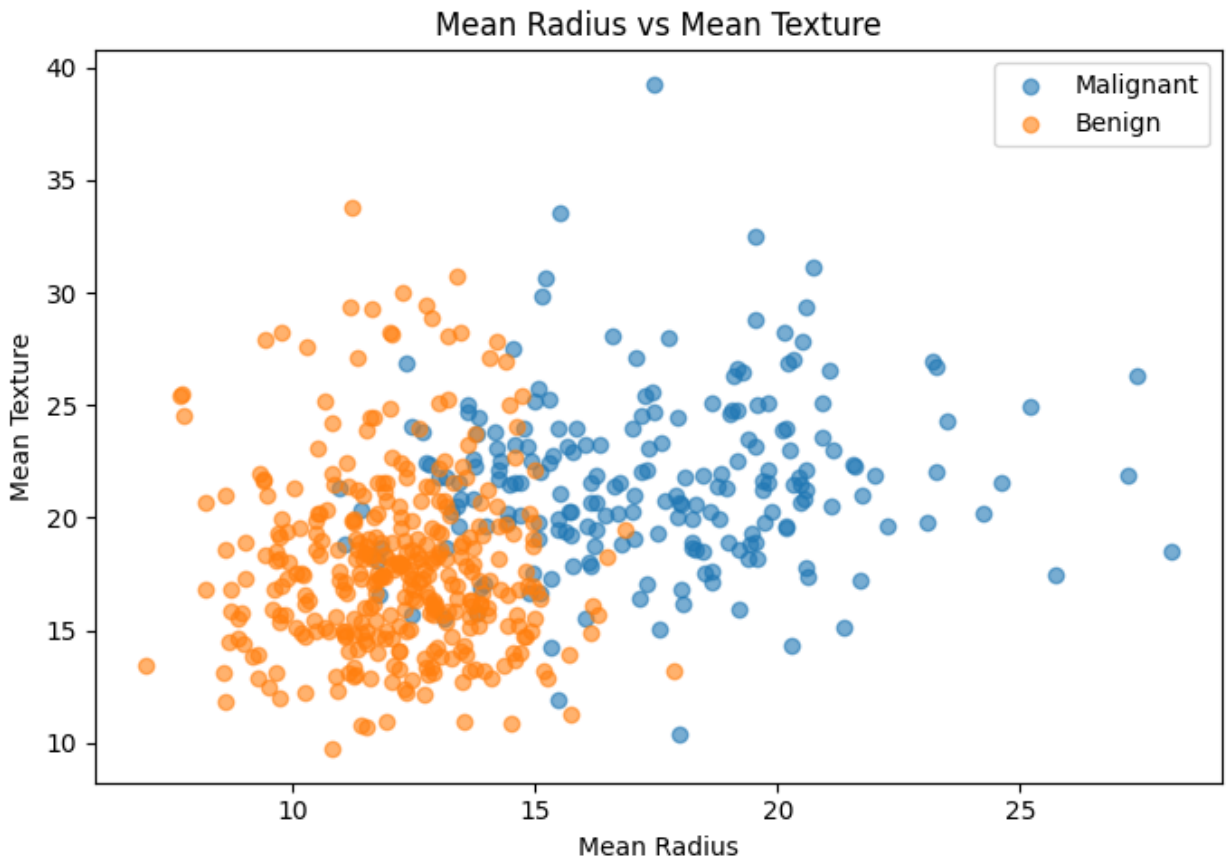


A moderately balanced dataset is indicated by the distribution of classes, which reveals 357 benign cases and 212 malignant instances. For classification tasks, balanced datasets are crucial because they help avoid biased predictions (Powers, 2020).



One significant finding is that there are no missing values in the dataset, which makes preprocessing easier and enables a more targeted analysis.

Before using Logistic Regression, feature scaling is required because the dataset's features differ in scale. According to Kuhn and Johnson (2013), standardization guarantees that every feature contributes equally to the model and avoids bias toward features with greater magnitudes.



Because the classes in this dataset are largely separable, the model may attain high performance without sacrificing interpretability, making it an excellent choice for illustrating Logistic Regression.

4. Methodology

Data preparation, feature scaling, model training, and evaluation are the main steps in the methodology used in this lesson.

First, a DataFrame is used to load the dataset and transform it into a structured representation. The goal variable (y), which stands for the class labels, is separated from the input characteristics (X).

Next, an 80–20 split is used to divide the dataset into training and testing sets. The model is trained using the training set, and its performance on untested data is assessed using the testing set. To maintain the class distribution in both sets, stratified sampling is used.

StandardScaler is used for feature scaling, standardizing the characteristics by scaling to unit variance and eliminating the mean. Because it enhances convergence and guarantees that every feature is handled equally, this step is crucial for logistic regression (Kuhn and Johnson, 2013).

The scaled training data is then used to train the Logistic Regression model. By optimizing its coefficients, the model discovers the connection between the goal variable and the input features (Hosmer et al., 2013).

Predictions are created using the test data following training. Apart from class predictions, probability estimates are also produced and utilized to assess the model using metrics like ROC-AUC (Fawcett, 2006).

Accuracy, precision, recall, and F1-score are all calculated as part of the evaluation procedure. Particularly in classification jobs where different kinds of errors have distinct consequences, these measures offer a thorough picture of the model's performance (Powers, 2020).

To further comprehend the data, visualizations like ROC curves and confusion matrices are produced. These visual aids aid in comprehending the model's ability to differentiate across classes.

5. Results and Analysis

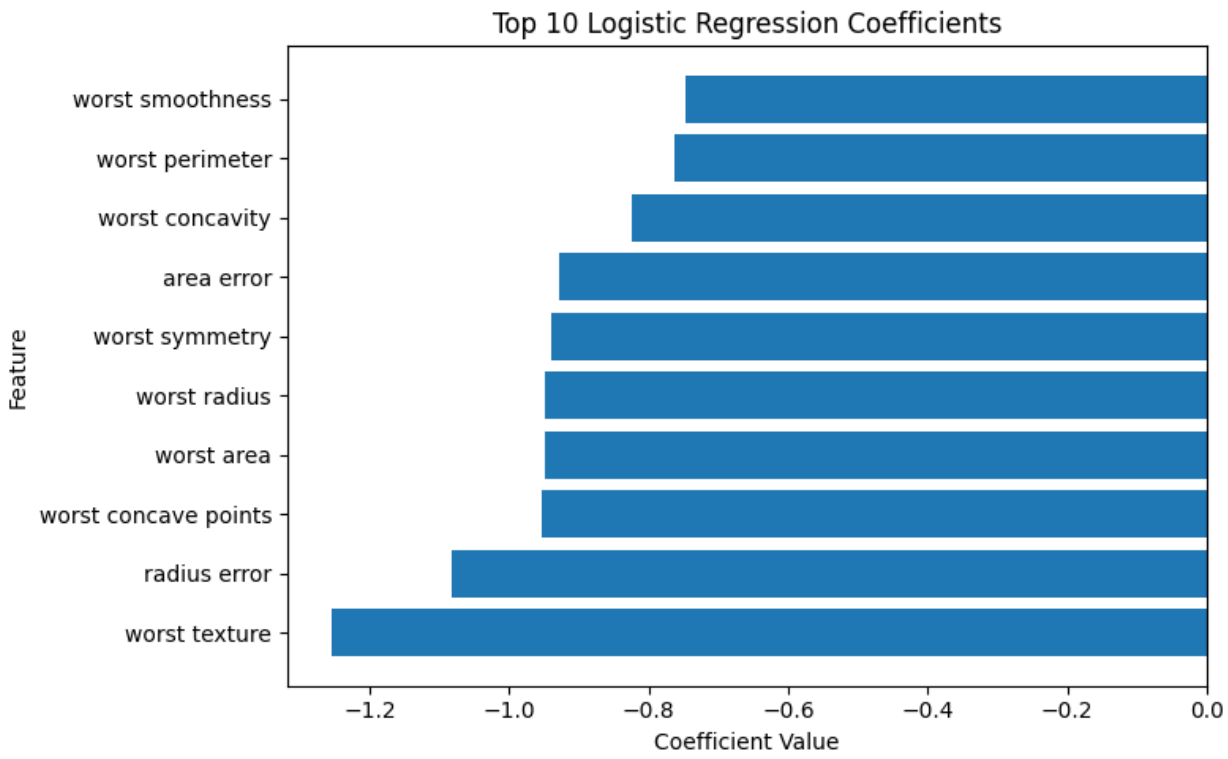
Strong classification capabilities were demonstrated by the Logistic Regression model's exceptionally high performance across all evaluation metrics.

The model's accuracy was 98.25%, indicating that most of the predictions were accurate. Although accuracy is a helpful indicator, performance in classification tasks is not entirely captured by it, particularly when there is class imbalance (Powers, 2020). The model does a good job of detecting both positive and negative situations, as evidenced by the precision and recall values of 98.61%.

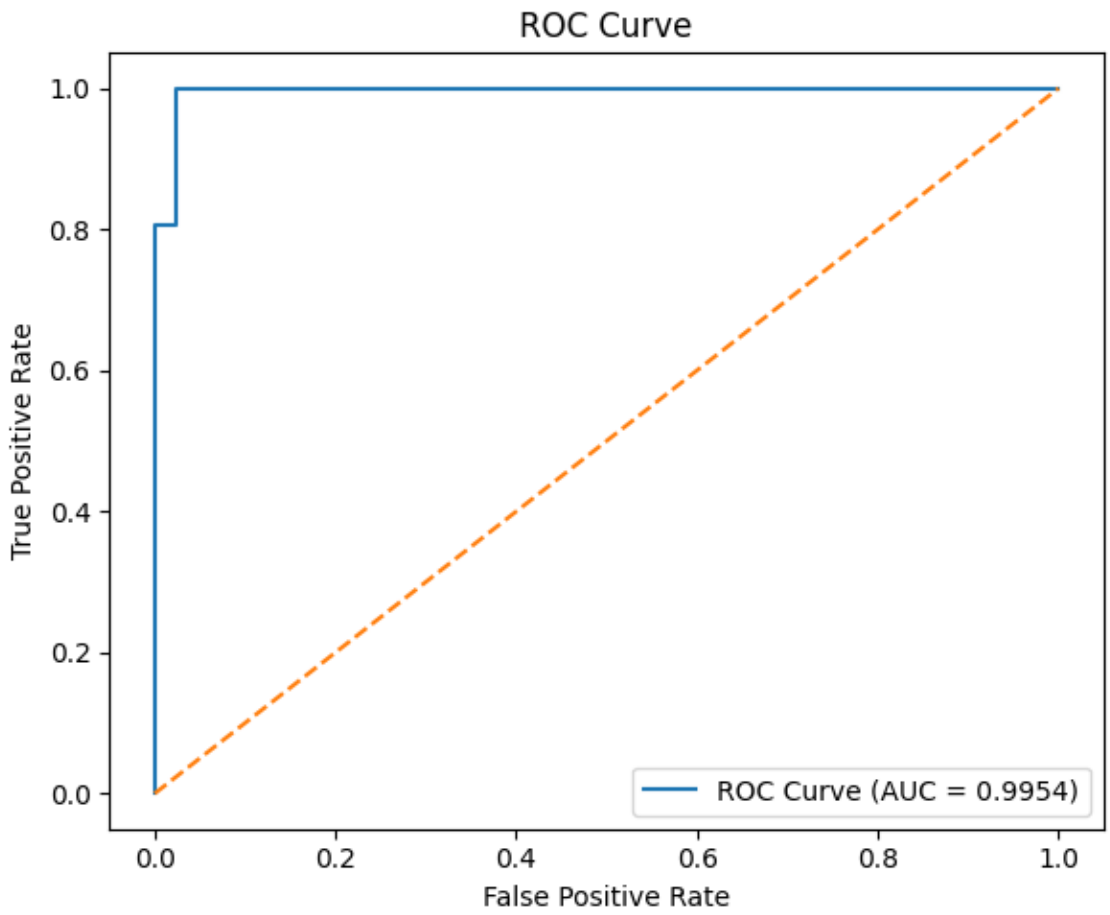
```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob)

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-score :", round(f1, 4))
print("ROC-AUC  :", round(auc, 4))
```

```
Accuracy : 0.9825
Precision: 0.9861
Recall   : 0.9861
F1-score : 0.9861
ROC-AUC  : 0.9954
```

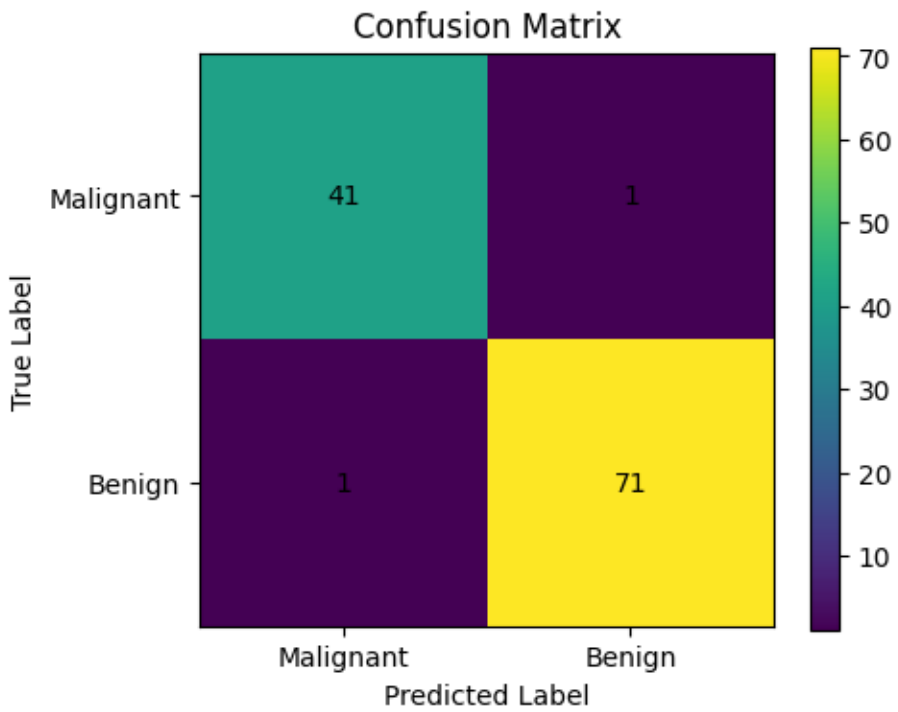
High precision indicates that the model is very likely to be accurate when it forecasts a tumor to be benign. A high recall means that the majority of real benign situations are successfully identified by the model. The model's performance was consistent, as evidenced by the F1-score of 98.61%, which strikes a compromise between precision and recall (Powers, 2020).



The ROC-AUC score was 0.9954, which is incredibly near to 1. This suggests that the model is quite good at differentiating between benign and malignant instances. approaching-perfect categorization ability is indicated by a number approaching 1 (Fawcett, 2006).



Only a very tiny percentage of samples were incorrectly classified, according to the confusion matrix. This indicates that the model is quite dependable for this particular dataset.



This conclusion is further supported by the ROC curve, which displays a curve that is extremely near to the top-left corner, indicating excellent sensitivity and specificity.

Features associated with tumor size and texture, such as mean radius and worst texture, had the biggest impact on the model's predictions, according to feature importance analysis. Given that these traits are important markers of cancer, this is consistent with medical understanding (Wolberg et al., 1995).

Overall, the findings show that the strong linear separability of the characteristics makes Logistic Regression quite effective for this dataset.

6. Critical Evaluation

Despite the model's outstanding performance, it is crucial to assess its advantages and disadvantages.

The ease of use and interpretability of logistic regression is one of its key advantages. In fields like healthcare, where explainability is crucial, the model offers lucid insights into how each feature contributes to the prediction (Hosmer et al., 2013).

Large-scale applications can benefit from the model's ease of implementation and computing efficiency (Raschka and Mirjalili, 2019).

Logistic regression does, however, have a number of drawbacks. It makes the assumption that features, and the target variable's log-odds have a linear relationship, which could not be true for complicated datasets (Bishop, 2006).

Its sensitivity to feature scaling is another drawback. The model might not converge effectively or yield erroneous findings if it is not scaled appropriately (Kuhn and Johnson, 2013).

The exceptionally high performance seen in this tutorial may possibly suggest that there is some overfitting or that the dataset is comparatively simple to classify. Real-world situations frequently involve more complicated data, and performance could be lower.

Additionally, highly correlated characteristics may be difficult for Logistic Regression to handle, which may have an impact on the coefficients' stability.

Regularization and cross-validation are two methods that can be used to enhance the model. Further insights can be obtained by contrasting Logistic Regression with more intricate models like Random Forest or Support Vector Machines (Breiman, 2001; Cortes and Vapnik, 1995).

7. Conclusion

Using the Breast Cancer Wisconsin dataset, this course illustrated how to use logistic regression for binary classification. The model performed exceptionally well on a number of evaluation criteria, including as ROC-AUC, accuracy, precision, and recall.

The findings show that datasets with linearly separable features benefit greatly from Logistic Regression. It is a useful technique in machine learning due to its efficiency, interpretability, and simplicity (James et al., 2021).

Its assumptions of linearity and sensitivity to feature scaling, in particular, should be taken into account. Even while it works well on this dataset, more intricate models could be needed for non-linear relationship problems in the real world.

All things considered, this lesson offers a thorough grasp of logistic regression and illustrates how it may be used to solve classification issues.

8. References

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